Category:

Network Forensic

Name:

Reversing & Packet Analysis

Message:

You are provided with process memory dump of a backdoor and packet capture data that includes flag value. Try reversing the backdoor to identify flag from the packet capture data.
https://cyberseagame2025.s3.us-east-2.amazonaws.com/Resources/csg_reversing_network_forensic_challenge.zip

Objective:

Identify hidden message (flag) from pcap file and decrypt with information extracted from binary file.

Instructions:

The pcap file contains telnet communication between client and backdoor carrying following messages.

- type c:¥temp¥flag.txt
- 1KeGOcMvvKpPbxrNRdyqJmSGh3j+ZCvIT5RKk6lTBkSGBYSYrX7dy3Q/E8wVuQAXNFMs978T1A4qyx7EeX9fvg==

“1KeGOcMvvKpPbxrNRdyqJmSGh3j+ZCvIT5RKk6lTBkSGBYSYrX7dy3Q/E8wVuQAXNFMs978T1A4qyx7EeX9fvg==” is the encrypted version of flag value.
You will need to reverse engineer binary file to acquire following information to decrypt it.

1. Encryption Key
- string advancedEncodedKey = "2a5b05310e08081f2b3405031134372a073c1f1c371a2e1c09303b4308393219";
2. Initial Vector (IV)
- string advancedEncodedIv = "252e0a0a3a1f291c1b343b39212b3c11013d312f08315158";
3. Function that decodes the Key and IV
4. Function that Encrypts the flag

Once 2 functions are identified, reverse engineering AES key and IV would get you the flag value.

- Decoded AES Key: rA8fT6YzX2pQ4eM9vW1kLdB3
- Decoded IV: uJ7nP0xR3aH5bG2q

- Decrypted Flag value: R3v3rsing_@nd_N3tw0rk_F0r3nsic_Ch@ll3ng3_S0lv3d!!